

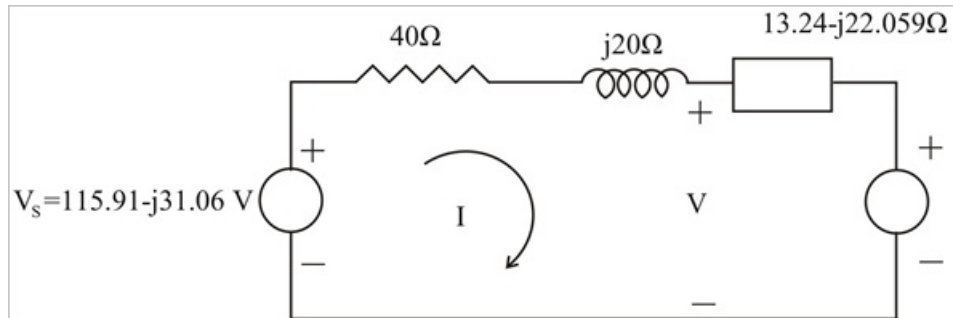
Problem

Rework Prob. 10.7 using source transformation.

Step-by-step solution

Step 1 of 2

$$50 \parallel (-j30) = \frac{50 \times (-j30)}{50 - j30} = 13.24 - j22.059$$



Step 2 of 2

Applying KVL gives,

$$-115.91 + j31.058 + (53.24 - j2.059)I - 134.95 + j74.912 = 0$$

$$\text{Or, } I = \frac{-250.86 + j105.97}{53.24 - j2.059} = -4.7817 + j1.8055$$

But,

$$-V_s + (40 + j20)I + V = 0$$

$$\Rightarrow V = V_s - (40 + j20)I$$

$$\begin{aligned} V &= 115.91 - j31.05 - (40 + j20)(-4.7817 + j1.8055) \\ &= 124.06 \angle -154^\circ \text{ V} \end{aligned}$$

Which agree with the result in problem 10.7